

Chapter: Test of Hypothesis

Question No. 1

a)

A soft drink company claims that the average amount of drink in its bottle is 500 ml. A random sample of 36 bottles shows an average amount of 492 ml. The population standard deviation is known to be 24 ml.

Requirement:

At the 0.05 significance level, test whether the average amount of drink is less than 500 ml.

b)

A bank wants to compare the average monthly savings of customers using mobile banking and customers using traditional banking. A sample of 45 mobile banking users shows an average monthly saving of BDT 8,200, with a population standard deviation of BDT 1,500. A sample of 50 traditional banking users shows an average monthly saving of BDT 7,600, with a population standard deviation of BDT 1,800.

Requirement:

At the 0.05 significance level, can we conclude that mobile banking users save more on average?

Chapter: Test of Hypothesis

Question No. 2

a)

A restaurant claims that the average waiting time for customers is 15 minutes. A sample of 25 customers shows an average waiting time of 17 minutes, with a sample standard deviation of 4 minutes.

Requirement:

At the 0.05 significance level, test whether the average waiting time is greater than 15 minutes.

b)

A training institute claims that students score an average of 75 marks after completing its preparation course. A sample of 40 students shows an average score of 72 marks. The population standard deviation is 8 marks.

Requirement:

At the 0.05 level of significance, can we conclude that the average score is less than 75 marks?

Chapter: Sampling Distribution and Estimation

Question No. 3

a)

A mobile operator is planning to launch a new internet package. A survey of 600 customers shows that 390 customers are interested in purchasing the package.

Requirement:

i. Estimate the population proportion of customers interested in the package.

ii. Construct a 95% confidence interval for the population proportion.

iii. Interpret the result.

b)

The manager of a garment factory wants to estimate the average daily output per worker. A sample of 30 workers shows an average output of 48 pieces per day, with a sample standard deviation of 6 pieces.

Requirement:

i. What is the best estimate of the population mean?

- ii. Which distribution should be used for constructing the confidence interval? Explain briefly.
- iii. For a 95% confidence interval, identify the appropriate test statistic.
- iv. Construct the 95% confidence interval for the population mean.
- v. Would it be reasonable to conclude that the population mean is 50 pieces? What about 60 pieces?

Chapter: Sampling Distribution and Estimation

Question No. 4

a)

A poultry farm owner wants to estimate the average monthly feed consumption per chicken. A sample of 20 chickens shows an average consumption of 4.5 kg, with a sample standard deviation of 0.8 kg.

Requirement:

- i. What is the best estimate of the population mean?
- ii. Which distribution should be used for this estimation problem?
- iii. What assumption is required?
- iv. Construct a 99% confidence interval for the population mean.
- v. Would it be reasonable to conclude that the population mean is 4 kg? What about 6 kg?

b)

A supermarket is considering introducing a new loyalty card. A random sample of 500 customers shows that 340 customers are willing to use the loyalty card.

Requirement:

- i. Estimate the population proportion.
- ii. Construct a 95% confidence interval for the population proportion.
- iii. Interpret your findings in the context of the problem.

Chapter: Correlation Analysis

Question No. 5

a)

The correlation coefficient ranges from -1.00 to $+1.00$. Explain this range graphically and interpret positive, negative, and zero correlation.

b)

Two supervisors rated the performance of 8 employees using an ordinal scale: Excellent, Good, Average, Fair.

Employee	Supervisor A Rating	Supervisor B Rating
1	Excellent	Good
2	Good	Good
3	Average	Average
4	Excellent	Excellent
5	Fair	Average
6	Good	Fair
7	Average	Good
8	Fair	Fair

Requirement:

Calculate the Spearman rank correlation coefficient. Do the two supervisors agree in their ratings?

Chapter: Regression Analysis

Question No. 6

The following data show annual sales and advertising expenditure of a company for 10 years.

Year	Sales (BDT Lakhs)	Advertising Expenditure (BDT Thousands)
2016	60	15
2017	75	20
2018	80	25
2019	90	30
2020	95	35
2021	105	40
2022	110	45
2023	125	50
2024	130	55
2025	140	60

Requirement:

- Draw a scatter diagram and interpret the relationship between advertising expenditure and sales.
- Formulate and interpret a two-variable regression model to estimate the effect of advertising expenditure on sales. Also estimate sales if advertising expenditure is BDT 75 thousand.

Chapter: Probability

Question No. 7

a)

A factory has three machines: A, B, and C. Machine A produces 30%, Machine B produces 45%, and Machine C produces 25% of the total output. The defective rates are 2%, 3%, and 5% respectively.

Requirement:

- Draw a tree diagram.
- Find the probability that a randomly selected product is defective.
- If a product is defective, find the probability that it came from Machine C.

b)

The weight of packets of rice produced by a company is normally distributed with a mean of 5 kg and a standard deviation of 0.12 kg. A packet is acceptable if its weight is between 4.80 kg and 5.20 kg.

Requirement:

Find the probability that a randomly selected packet is acceptable.

Chapter: Probability

Question No. 8

a)

A manufacturer knows that 3% of its bulbs are defective. If an inspector randomly selects 12 bulbs, find the probability that at least one bulb is defective.

b)

In a survey of 200 students, 70% use Facebook, 50% use Instagram, and 30% use both. If a student is selected randomly and it is known that the student uses Facebook, find the probability that the student also uses Instagram.

c)

The lifetime of batteries produced by a company is normally distributed with a mean of 1,200 hours and a standard deviation of 80 hours. A battery is considered standard if its lifetime is between 1,100 hours and 1,350 hours.

Requirement:

Calculate the probability that a randomly selected battery is standard.

Chapter: Measures of Central Tendency and Measures of Dispersion

Question No. 9

a)

A company surveyed small retail shops to find the number of customers served per day. The frequency distribution is given below:

Number of Customers	Frequency
20 up to 30	4
30 up to 40	10
40 up to 50	18
50 up to 60	12
60 up to 70	6

Requirement:

Estimate the mean, standard deviation, and coefficient of variation.

b)

The following frequency distribution shows the delivery time of online orders:

Delivery Time in Days	Frequency
1–3	5
3–5	9
5–7	16
7–9	12
9–11	6
11–13	2

Requirement:

Construct a box plot for the above data. Comment on the shape of the distribution.

Chapter: Measures of Central Tendency and Measures of Dispersion

Question No. 10

a)

The following table shows the monthly expenditure of a sample of households:

Monthly Expenditure, BDT Thousand No. of Households

10–20	8
20–30	16
30–40	30
40–50	24
50–60	14
60–70	8

Requirement:

Calculate the coefficient of skewness and interpret the value.

b)

The following frequency distribution gives the weights of packets produced by a machine:

Weight in Grams	Frequency
90–100	3
100–110	8
110–120	14
120–130	16
130–140	7
140–150	2

Requirement:

Construct a box plot and comment on the shape of the distribution.

Chapter: Introduction to Business Statistics, Collection of Data and Presentation of Data

Question No. 11

a)

A pie chart shows the types of transport used by 360 students to come to university. The percentages are as follows:

Transport Type	Percentage
Bus	40%
Rickshaw	25%
Bicycle	15%
Walking	20%

Requirement:

- Calculate the number of students using buses.
- Calculate the number of students using rickshaws.
- Calculate the number of students who walk to university.

b)

Identify the measurement scale: Nominal, Ordinal, Interval, or Ratio.

- Blood group of patients
- Temperature in Celsius
- Monthly income of employees
- Ranking of students in a class
- Brand names of mobile phones
- Number of children in a family
- Satisfaction level: Very satisfied, Satisfied, Neutral, Dissatisfied
- Calendar years
- Weight of rice bags
- National ID numbers

c)

A cumulative frequency polygon shows the number of customers according to monthly purchase amount.

Requirement:

- What is the class interval?
- What is the total number of customers studied?
- How many customers spent less than a given boundary value?
- How many customers spent between two given boundary values?

Chapter: Introduction to Business Statistics, Collection of Data and Presentation of Data

Question No. 12

a)

Write YES/NO depending on which measure of central tendency can be used for the following levels of measurement.

Level of Measurement Mode Median Mean

Nominal

Ordinal

Interval/Ratio

b)

Identify the measurement scale: Nominal, Ordinal, Interval, or Ratio.

i. Customer feedback: Poor, Average, Good, Excellent

ii. Types of bank accounts

iii. Time shown on a digital clock

iv. Distance from home to workplace

v. Examination scores

vi. Shirt sizes: Small, Medium, Large

vii. Product serial numbers

viii. Age of respondents

ix. Political party preference

x. Year of birth

c)

The following table shows the number of hours students spent watching online lectures in one week:

Hours Spent	Frequency
0–2	4
2–4	10
4–6	20
6–8	12
8–10	4

Requirement:

Construct a histogram and draw a frequency polygon from it.

Short Questions

Question No. 13

Answer any ten of the following questions.

a) Define statistics.

b) What is the difference between primary data and secondary data?

c) What is meant by population and sample?

d) Define parameter and statistic.

e) What is a sampling distribution?

f) What is the standard error of the mean?

g) What is a confidence interval?

h) What is the meaning of 95% confidence level?

i) What is a null hypothesis?

- j) What is an alternative hypothesis?
- k) What is Type I error?
- l) What is Type II error?
- m) What is p-value?
- n) What is level of significance?
- o) What is meant by one-tailed test and two-tailed test?

Question No. 14

Answer any ten of the following questions.

- a) Define probability.
- b) What is conditional probability?
- c) State the addition rule of probability.
- d) State the multiplication rule of probability.
- e) What is Bayes' theorem?
- f) What is a normal distribution?
- g) Mention two properties of normal distribution.
- h) What is z-score?
- i) What is correlation?
- j) What is the range of the correlation coefficient?
- k) What does a negative correlation mean?
- l) What is Spearman's rank correlation?
- m) What is regression analysis?
- n) What is the difference between dependent and independent variable?
- o) What is a scatter diagram?

Question No. 15

Answer any ten of the following questions.

- a) Define mean, median, and mode.
- b) What is range?
- c) What is variance?
- d) What is standard deviation?
- e) What is coefficient of variation?
- f) What is skewness?
- g) What is a positively skewed distribution?
- h) What is a negatively skewed distribution?
- i) What is a box plot?
- j) What are quartiles?
- k) What is interquartile range?
- l) What is a histogram?
- m) What is a frequency polygon?
- n) What is an ogive?
- o) What is the difference between nominal and ordinal scale?